Eta invariants of transverse-traceless tensor curl on three-dimensional lens spaces

Luqman Mahmood*

30 September 2026

Abstract

Let $L(p,q)$ be a round lens space with $p \geq 2$ and let A_J be the tensor-curl operator obtained by compressing Homma's normalized three-dimensional self-gradient to divergence-free trace-free symmetric tensors of rank $J \geq 1$. We show that A_J is a self-adjoint projected elliptic pseudodifferential operator with compact resolvent, and we evaluate its eta invariant for every unit q modulo p. The result is an exact finite expression combining a classical Dedekind sum, elementary signed residue sums, finite Fourier-aliasing corrections, and an explicit low-mode count. For homogeneous quotients it reduces to

$$\eta_J(L(p,1)) = -\frac{J\big(p^2 - 3Jp + 4J^2 - 2\big)}{3p}, \qquad p \geq \max(2, 2J-1).$$

The threshold is sharp. We further establish rank periodicity and the reflection $\eta_{p-J} = -\eta_J$ when $1 \pm q$ are units modulo p, and we exhibit four parameterized eta-zero families at $p = 2J^2 - 1$ (with possible coincidences at the smallest ranks). At $J = 1$ the formula recovers the classical odd-signature value $-4s(q,p)$ and matches the normalization in Jeffrey's Chern–Simons spectral-flow formula; in rank two it evaluates the spectral-asymmetry (eta) contribution to the determinant phase set aside by Castro, Lashkari, and Maloney, not a complete one-loop phase. The operator, its sphere spectrum, and the equivariant character method are classical; the contribution is the explicit signed evaluation, uniform in the rank and in prime and composite levels. The sphere S^3 (the $p = 1$ case) has symmetric TT spectrum and eta invariant zero.

1 Introduction

Spectral determinants on spherical quotients often retain the magnitude of a first-order spectrum while discarding its sign. For transverse-traceless (TT) tensors this loses precisely the spectral asymmetry measured by an eta invariant. The arbitrary-rank TT representations and quotient character machinery are classical in the heat-kernel literature [DGG10]. In rank two, Castro–Lashkari–Maloney write down the same tensor-curl operator and its lens space degeneracies, but explicitly set aside the sign or determinant phase because only the norm is used [CLM11, eqs. (3.3), (3.13)–(3.18) and footnote 5]. This paper evaluates that signed difference and its integer-rank extension.

The local starting point is Homma's three-dimensional higher-spin self-gradient [Hom01, HT21]. It is not elliptic on the unrestricted trace-free symmetric-tensor bundle. Its compression to the

*Copyright 2026 Luqman Mahmood. This paper is licensed under Creative Commons Attribution 4.0 International (CC BY 4.0); see https://creativecommons.org/licenses/by/4.0/. Software is excluded from this grant.

solenoidal range, however, has principal symbol $\pm|\xi|$. We formulate this compression as a projected, or Toeplitz, elliptic operator. On a round lens quotient its two chiral branches are

$$V_{n+2J} \otimes V_n, \qquad V_n \otimes V_{n+2J}, \qquad n \geq 0,$$

with signed eigenvalues $\pm(n + J + 1)$ on the unit three-sphere. Here V_a denotes the irreducible $SU(2)$ representation of highest weight a and dimension $a + 1$.

Our main result is a closed formula for the eta invariant after cyclic projection. It applies at every level $p \geq 2$ and explicitly retains both removable denominator-singular character limits and the modular aliasing terms that are easy to lose in a generic trigonometric calculation. At even order the half-turn is the sole additional denominator-singular class; its two chiral TT character products agree termwise, so its signed contribution is zero before analytic continuation. Three consequences are particularly compact:

(i) the homogeneous polynomial in p and J displayed in the abstract;

(ii) rank periodicity and antisymmetry on nonsingular lens rows; and

(iii) four infinite arithmetic zero families at $p = 2J^2 - 1$.

At $J = 1$ the result is the established curl or odd-signature eta invariant [APS75, Bär19]. This fixes the normalization. A second consistency check comes from the appearance of the same $J = 1$ term in Jeffrey's $SU(2)$ Chern–Simons spectral-flow formula [Jef92]; Appendix B records the comparison.

The operator, sphere spectrum, quotient projection, and equivariant eta methods are antecedent machinery. To the best of our knowledge, the explicit signed all-rank lens formula and its uniform treatment of composite levels have not appeared before. The result is a spectral statement for round lens spaces; completing a gravitational or higher-spin determinant would require additional sectors and choices not considered here.

Section 2 fixes the quotient and orientation conventions. Sections 3 and 4 construct the projected operator and continue its eta series. Sections 5–7 prove the closed formula and its rank-symmetry and zero-family consequences. The final sections record computational checks, place the result among its antecedents, and discuss open problems.

2 Lens-space and orientation conventions

Write the unit three-sphere as

$$S^3 = \{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 = 1\}.$$

For $p \geq 2$ and $q \in (\mathbb{Z}/p\mathbb{Z})^\times$, the generator of $\mathbb{Z}_p$ acts by

$$(z_1, z_2) \longmapsto \big(\mathrm{e}^{2\pi i/p} z_1, \mathrm{e}^{2\pi i q/p} z_2\big),$$

and $L(p, q)$ denotes the quotient with its complex quotient orientation. We choose the branch $V_{n+2J} \otimes V_n$ to have positive eigenvalue. Changing either global convention reverses every eta value simultaneously. Throughout, η means the ordinary eta invariant of A_J, not the reduced eta $(\eta+h)/2$; the zero-mode convention is therefore kept separate. The round-boundary completion in [Mah26d] uses this same positive-left-heavy TT convention; its auxiliary finite character is subtracted when forming the full boundary eta invariant. The case $p = 1$ is the sphere S^3, whose two TT branches

have equal multiplicity at every level and hence eta invariant zero; it is not included in the nontrivial lens-space theorem below.

Let $q^* \in \{1, \dots, p-1\}$ be the least positive inverse of q modulo p. For an integer k, let $\langle k \rangle_p \in \{0, \dots, p-1\}$ be its least nonnegative residue and define

$$\rho_p(k) = \begin{cases} 0, & p \mid k, \\ p - 2\langle k \rangle_p, & p \nmid k, \end{cases} \qquad \delta_p(k) = \mathbf{1}_{p|k}. \tag{1}$$

The sawtooth function and Dedekind sum are

$$((x)) = \begin{cases} x - \lfloor x \rfloor - \frac{1}{2}, & x \notin \mathbb{Z}, \\ 0, & x \in \mathbb{Z}, \end{cases} \qquad s(q,p) = \sum_{a=0}^{p-1} \left(\!\left(\frac{a}{p}\right)\!\right) \left(\!\left(\frac{qa}{p}\right)\!\right).$$

For $d \geq 0$, set $W_d = \{-d, -d+2, \dots, d\}$. For $a, b \geq 1$ with $a \equiv b \pmod 2$, the cyclic invariant count in $V_{a-1} \otimes V_{b-1}$ is

$$I_{a,b}(p,q) = \#\left\{(r,t) \in W_{a-1} \times W_{b-1} : (1+q)r + (1-q)t \equiv 0 \pmod{2p}\right\}. \tag{2}$$

Finally, define the finite low-mode correction

$$L_J(p,q) = \sum_{m=1}^{J-1} \left[I_{J-m,J+m}(p,q) - I_{J+m,J-m}(p,q) \right]. \tag{3}$$

The sum is empty at $J = 1$.

symbol	meaning
A_J	normalized rank-J self-gradient compressed to TT tensors
$\rho_p(k)$	signed least-residue function in (1)
$\delta_p(k)$	divisibility indicator $\mathbf{1}_{p\|k}$
$I_{a,b}(p,q)$	cyclic invariant count in $V_{a-1} \otimes V_{b-1}$
$L_J(p,q)$	finite subtraction of formal modes below the TT threshold

Remark 2.1 (Radius scaling)**.** If the round metric is rescaled from unit radius to radius R, then A_J and every nonzero eigenvalue scale by R^{-1}. Consequently $\eta_{A_J,R}(s) = R^s \eta_{A_J,1}(s)$, and the value $\eta_{A_J,R}(0)$ in the theorem below is independent of R.

The quotient classification symmetries give, in these conventions,

$$\eta_J(L(p,q^*)) = \eta_J(L(p,q)), \qquad \eta_J(L(p,-q)) = -\eta_J(L(p,q)). \tag{4}$$

Indeed, swapping the two complex coordinates and then replacing the cyclic generator by its q^*th power gives the orientation-preserving $q \mapsto q^*$ identification. Conjugating the second complex coordinate gives the orientation-reversing $q \mapsto -q$ identification; orientation reversal negates tensor curl and hence its eta invariant.

3 The projected TT tensor-curl operator

Let (M, g) be a closed oriented constant-curvature three-manifold. We work on the complexification of the real tensor bundle and put

$$E_J = \mathrm{Sym}_0^J T^*M.$$

Let D^0_{2J} be Homma's self-gradient on the even-spin bundle identified with E_J, let

$$U_J = \frac{D^0_{2J}}{2J},$$

and let $\delta_J : E_J \to E_{J-1}$ be trace-free symmetric divergence. If P_J is the exact L^2-orthogonal projection onto $\ker \delta_J$, define

$$A_J = P_J U_J P_J : P_J H^1(E_J) \subset P_J L^2(E_J) \longrightarrow P_J L^2(E_J). \tag{5}$$

Theorem 3.1 (Projected TT operator)**.** *The operator A_J is self-adjoint and has compact resolvent. Its compressed principal symbol on $\operatorname{ran}\sigma(P_J)(\xi)$ has eigenvalues $+|\xi|$ and $-|\xi|$ for every $\xi \neq 0$. Hence A_J is elliptic in the projected pseudodifferential calculus. On the unit round sphere its TT spectrum is*

$$\begin{aligned} &V_{n+2J} \otimes V_n : +(n+J+1), \\ &V_n \otimes V_{n+2J} : -(n+J+1), \qquad n \geq 0. \end{aligned} \tag{6}$$

The spectrum on a round spherical quotient is obtained by taking invariant vectors in each representation. In particular, there are no zero modes.

Proof. Write

$$d(\xi) = \sigma_1(\delta_J)(x,\xi), \qquad u(\xi) = \sigma_1(U_J)(x,\xi).$$

The even spin module V_{2J} factors through $SO(3)$ and identifies with the fiber of $\operatorname{Sym}^J_0 T^*M$. After rotating a nonzero covector ξ to a fixed axis, its stabilizer decomposes that fiber into one-dimensional weight spaces

$$E_{J,x} = \bigoplus_{m=-J}^{J} \mathbb{C} e_m.$$

Homma's explicit symbol formulae, in the Hermitian Fourier convention and with $U_J = D^0_{2J}/(2J)$, give

$$u(\xi)e_m = \frac{m}{J}|\xi| e_m, \qquad \|d(\xi)e_m\|^2 = |\xi|^2 \frac{J^2 - m^2}{J(2J-1)}. \tag{7}$$

The target $E_{J-1,x}$ has precisely the weights $-(J-1), \dots, J-1$. Thus $d(\xi)$ is nonzero on every source line that maps to a target weight, vanishes exactly on the two extreme lines, and is surjective. Consequently

$$\ker d(\xi) = \operatorname{span}\{e_{-J}, e_J\}, \qquad u(\xi)|_{\ker d(\xi)} = \operatorname{diag}(-|\xi|, +|\xi|),$$

up to the single global sign fixed in Section 2.

We next construct the exact global projection. Put

$$\Delta_{\delta,J} = \delta_J \delta_J^*, \qquad K_{J-1} = \ker \delta_J^*.$$

Surjectivity of $d(\xi)$ implies that $\Delta_{\delta,J}$ is elliptic of order two. It is nonnegative and $\ker \Delta_{\delta,J} = K_{J-1}$. Hence K_{J-1} is finite dimensional and smooth, and its orthogonal projector $\Pi_{K_{J-1}}$ is smoothing. The positive operator $\Delta_{\delta,J} + \Pi_{K_{J-1}}$ is invertible, with

$$G_J = (\Delta_{\delta,J} + \Pi_{K_{J-1}})^{-1} \in \Psi^{-2}, \qquad \Delta_{\delta,J} G_J = G_J \Delta_{\delta,J} = I - \Pi_{K_{J-1}}.$$

Since $\operatorname{ran}\delta_J \perp K_{J-1}$, $\Pi_{K_{J-1}} \delta_J = 0$. Therefore

$$P_J = I - \delta_J^* G_J \delta_J$$

satisfies

$$\delta_J P_J = \delta_J - \Delta_{\delta,J} G_J \delta_J = \delta_J - (I - \Pi_{K_{J-1}})\delta_J = 0.$$

It is self-adjoint and acts as the identity on $\ker \delta_J$. It is therefore exactly the L^2-orthogonal projection onto that kernel. Its principal symbol is

$$p(\xi) = I - d(\xi)^*(d(\xi)d(\xi)^*)^{-1}d(\xi), \tag{8}$$

the orthogonal projection onto $\ker d(\xi)$. In particular $P_J \in \Psi^0$ and is bounded on every Sobolev space; compare the standard solenoidal construction in [GL21].

The compressed principal symbol of $A_J = P_J U_J P_J$ is

$$p(\xi)u(\xi)p(\xi) : \operatorname{ran} p(\xi) \longrightarrow \operatorname{ran} p(\xi).$$

By (7), its eigenvalues are $\pm|\xi|$, so it is invertible for $\xi \neq 0$. This proves projected ellipticity [SS99, Sei15].

It remains to identify the closed domain. Let $Q_J = I - P_J$ and

$$\Lambda = (1 + \nabla^*\nabla)^{1/2}, \qquad \mathcal{B}_J = P_J U_J P_J + Q_J \Lambda Q_J.$$

The complex-power calculus gives $\Lambda \in \Psi^1$ with scalar principal symbol $|\xi|$. Because U_J is formally self-adjoint and P_J, Q_J are orthogonal projections, $\mathcal{B}_J$ is formally self-adjoint. Relative to

$$E_{J,x} = \operatorname{ran} p(\xi) \oplus \operatorname{ran}(I - p(\xi)),$$

its order-one symbol is block diagonal:

$$\sigma_1(\mathcal{B}_J)(\xi) = \begin{pmatrix} p(\xi)u(\xi)p(\xi)|_{\operatorname{ran} p(\xi)} & 0 \\ 0 & |\xi| I \end{pmatrix}.$$

Both blocks are invertible, hence $\mathcal{B}_J$ is an elliptic classical operator of order one.

Let $\mathcal{B}_{\min}$ be its graph closure from smooth sections and $\mathcal{B}_{\max}$ its distributional maximal realization. If $v \in L^2$ and $\mathcal{B}_J v \in L^2$, an order--1 parametrix C with $C\mathcal{B}_J = I - R$, R smoothing, gives

$$v = C\mathcal{B}_J v + Rv \in H^1.$$

Conversely, an order-one operator maps H^1 continuously to L^2, so $\mathcal{D}(\mathcal{B}_{\max}) = H^1(E_J)$. The elliptic estimate

$$\|v\|_{H^1} \leq C_0(\|\mathcal{B}_J v\|_{L^2} + \|v\|_{L^2})$$

and density of smooth sections in H^1 show that $\mathcal{D}(\mathcal{B}_{\min}) = H^1(E_J)$ as well. Formal symmetry now yields

$$\mathcal{B}_{\min}^* = \mathcal{B}_{\max} = \mathcal{B}_{\min};$$

thus $\mathcal{B}_J$ is self-adjoint on $H^1(E_J)$. For nonreal z, elliptic regularity makes $(\mathcal{B}_J - z)^{-1} : L^2 \to H^1$ bounded. Composing with the compact Rellich inclusion proves that its resolvent is compact.

Algebraically,

$$P_J \mathcal{B}_J = \mathcal{B}_J P_J = P_J U_J P_J.$$

Thus $\operatorname{ran} P_J$ is reducing. The corresponding restriction is A_J, with domain

$$H^1(E_J) \cap \operatorname{ran} P_J = P_J H^1(E_J).$$

It is self-adjoint, and its resolvent is the compact compression of the resolvent of $\mathcal{B}_J$.

Finally, Homma's lowering operator D^-_{2J} is a nonzero constant multiple of δ_J under the even-spin/STF identification. On constant curvature, Homma's intertwining identity reads

$$D^0_{2J-2}D^-_{2J} = D^-_{2J}D^0_{2J}.$$

Therefore U_J preserves $\ker \delta_J$, and the compression agrees there with the local self-gradient. On S^3 the solenoidal trace-free sector is

$$\bigoplus_{n\geq 0} \left[V_{n+2J} \otimes V_n \oplus V_n \otimes V_{n+2J}\right].$$

Homma's signed spectrum for D^0_{2J} is $+2J(n+J+1)$ on the first summand and $-2J(n+J+1)$ on the second [Hom01, HT21]. Dividing by $2J$ proves (6). If a finite group acts freely by orientation-preserving isometries, pullback identifies quotient sections with invariant sections upstairs and intertwines both δ_J and U_J. The quotient eigenspaces are therefore exactly the invariant vectors in the displayed modules. Their eigenvalue magnitudes are $n+J+1>0$, so no zero mode occurs. □

Remark 3.2. The theorem does not make U_J elliptic on all of E_J, and the range of P_J is not the section space of a local rank-two vector bundle. The transverse projection is nonlocal. The intrinsic projected operator alone is not asserted to be a local boundary operator on an arbitrary three-manifold. The separate round boundary interpretation is described in the next subsection and is not an assumption of the eta calculation.

Lemma 3.3 (Rank-one normalization)**.** *Under the metric identification $E_1 = T^*M \otimes \mathbb{C}$, the normalized self-gradient is*

$$U_1 = \frac{D^0_2}{2} = *d.$$

Moreover δ_1 is the codifferential up to the convention fixed above, so A_1 is the classical curl operator on coclosed one-forms. On a lens space this is the coexact sector, and in the orientation of Section 2,

$$\eta(A_1; L(p,q)) = -4s(q,p).$$

Proof. In an oriented orthonormal frame the unique $SO(3)$-equivariant self-gradient on one-forms has components $(U_1\alpha)_i = \varepsilon_i{}^{jk}\nabla_j\alpha_k$, which is $*d\alpha$. This fixes the operator normalization rather than only its spectrum. The lowering operator is contraction with ∇, hence a convention-fixed nonzero multiple of the codifferential; its kernel is the coclosed subspace. Because $H^1(L(p,q);\mathbb{R}) = 0$, coclosed one-forms have no harmonic summand and are coexact. The classical odd-signature/curl formula in the chosen quotient orientation is $-4s(q,p)$ [APS75, Bär19]. □

3.1 Relation to the round boundary operator

For comparison with the companion boundary problem, on a round quotient the rank-J TT sector is a reducing block on which the companion boundary operator restricts to A_J in the convention of Section 2. The square completion, twisted-Dirac identification, and assembly of the full Atiyah–Patodi–Singer boundary correction—which involves auxiliary eta and kernel terms beyond $\eta(A_J)$—are carried out in [Mah26d] and are not needed here. The rankwise Dirac-to-tensor-curl transfer and the generic McKay–Chebyshev algebra appear separately in [Mah26c, Mah26a].

4 Cyclic projection and eta continuation

From (2) and (6), put

$$d^+_{J,n} = I_{n+2J+1,n+1}(p,q), \qquad d^-_{J,n} = I_{n+1,n+2J+1}(p,q), \tag{9}$$

and $D_J(n;p,q) = d^+_{J,n} - d^-_{J,n}$. For $\Re s$ large, the eta function is

$$\eta_{A_J}(s) = \sum_{n\geq 0} \frac{D_J(n;p,q)}{(n+J+1)^s}. \tag{10}$$

Proposition 4.1 (Affine residue continuation)**.** *For each residue $0 \leq r < 2p$ there are integers $A_r = A_r(J,p,q)$ and $B_r = B_r(J,p,q)$ such that*

$$D_J(r+2pk;p,q) = A_r k + B_r \qquad (k \geq 0). \tag{11}$$

With $a_r = (r+J+1)/(2p)$,

$$\eta_{A_J}(s) = (2p)^{-s} \sum_{r=0}^{2p-1} \left[A_r \zeta(s-1,a_r) + (B_r - A_r a_r)\zeta(s,a_r)\right]. \tag{12}$$

In particular, the eta function is regular at $s=0$ and

$$\eta_{A_J}(0) = \sum_{r=0}^{2p-1} \left[A_r\left(\frac{a_r^2}{2} - \frac{1}{12}\right) + B_r\left(\frac{1}{2} - a_r\right)\right]. \tag{13}$$

Proof. For a highest weight $h \geq 0$ and $a \in \mathbb{Z}/(2p)\mathbb{Z}$, let

$$c_h(a) = \#\{w \in W_h : w \equiv a \pmod{2p}\}.$$

The interval of weights grows by one full residue period at each end when h is replaced by $h+2p$. Consequently

$$c_{h+2p}(a) = c_h(a) + 2\mathbf{1}_{a\equiv h \pmod 2}. \tag{14}$$

Introduce the bilinear invariant-counting form

$$\mathcal{I}_{p,q}(c,d) = \sum_{\substack{a,b \pmod{2p} \\ (1+q)a+(1-q)b\equiv 0 \pmod{2p}}} c(a)d(b).$$

Fix $0 \leq r < 2p$, put $c = c_{r+2J}$ and $d = c_r$, and let $u(a) = 2\mathbf{1}_{a\equiv r \pmod 2}$. The two highest weights have the same parity. Iterating (14) therefore gives

$$c_{r+2J+2pk} = c + ku, \qquad c_{r+2pk} = d + ku.$$

Hence

$$\begin{aligned} D_J(r+2pk;p,q) &= \mathcal{I}_{p,q}(c+ku, d+ku) - \mathcal{I}_{p,q}(d+ku, c+ku) \\ &= \mathcal{I}_{p,q}(c,d) - \mathcal{I}_{p,q}(d,c) \\ &\quad + k\big[\mathcal{I}_{p,q}(u,d) + \mathcal{I}_{p,q}(c,u) - \mathcal{I}_{p,q}(u,c) - \mathcal{I}_{p,q}(d,u)\big]. \end{aligned}$$

The two quadratic terms are identical and cancel. This is (11), with the constant and linear coefficients in the last display equal to B_r and A_r.

Splitting (10) into these residue classes, and writing $a_r = (r + J + 1)/(2p)$, gives for $\Re s$ large

$$\begin{aligned}\eta_{A_J}(s) &= (2p)^{-s} \sum_{r=0}^{2p-1} \sum_{k\geq 0} (A_r k + B_r)(k + a_r)^{-s} \\ &= (2p)^{-s} \sum_{r=0}^{2p-1} \left[A_r \zeta(s-1, a_r) + (B_r - A_r a_r)\zeta(s, a_r)\right].\end{aligned}$$

The right side is the meromorphic continuation (12) and is regular at zero. Finally,

$$\zeta(0, a) = \frac{1}{2} - a, \qquad \zeta(-1, a) = -\frac{1}{2}\left(a^2 - a + \frac{1}{6}\right),$$

and substitution gives (13). □

The explicit Hurwitz formula proves regularity for the reducing TT spectrum directly. It is consistent with the general residue criteria for eta functions of elliptic pseudodifferential operators [Pon06, SS13].

5 The all-rank lens-space formula

Theorem 5.1 (All-rank eta formula)**.** *For every integer $J \geq 1$, every $p \geq 2$, and every $q \in (\mathbb{Z}/p\mathbb{Z})^\times$, the standard eta invariant of (5) is*

$$\begin{aligned}\eta_J(L(p,q)) = &-4J\, s(q,p) \\ &+ \frac{2}{p} \sum_{a=1}^{J-1} (J-a)\left[\rho_p(aq) + \rho_p(aq^*)\right] \\ &- 2 \sum_{\substack{a,b\geq 1 \\ a+b\leq J-1}} (J-a-b)\left[\delta_p(aq-b) - \delta_p(aq+b)\right] \\ &- L_J(p,q),\end{aligned} \tag{15}$$

where (1) and (3) define every term.

The formula is valid without a parity or primality assumption. The third line records finite Fourier-aliasing congruences, including composite-level coincidences, and the fourth line removes the formal modes below the physical threshold $n + J + 1 = J + 1$.

5.1 Character reduction

For a nonidentity group element indexed by $1 \leq \ell < p$, set

$$\alpha_\ell = \frac{\pi \ell q}{p}, \qquad \beta_\ell = \frac{\pi \ell}{p}.$$

Also put

$$u_\ell = \alpha_\ell + \beta_\ell, \qquad v_\ell = \beta_\ell - \alpha_\ell,$$

and write

$$\chi_h(\theta) = \frac{\sin((h+1)\theta)}{\sin\theta} = \sum_{w\in W_h} e^{iw\theta}.$$

Extend the Weyl formula to integer indices by

$$\chi_{-1} = 0, \qquad \chi_{-h-2} = -\chi_h \quad (h \geq 0).$$

Lemma 5.2 (Central descent)**.** *The cyclic action on each TT branch and on every invariant count $I_{a,b}(p,q)$ in (2) is independent of the lift from $SO(4)$ to $Spin(4) = SU(2)_L \times SU(2)_R$, for both odd and even p.*

Proof. The two lifts differ by the diagonal central element $(-1,-1)$. On $V_{n+2J} \otimes V_n$ it acts by

$$(-1)^{(n+2J)+n} = (-1)^{2(n+J)} = 1,$$

and the exchanged TT branch has the same exponent. Likewise, $V_{a-1} \otimes V_{b-1}$ in (2) has central exponent $a+b-2$, which is even because $a \equiv b \pmod 2$. Thus all representations entering the signed projector descend to $SO(4)$ and the order-p cyclic average is well-defined without a parity assumption on p. □

For $a \equiv b \pmod 2$, character averaging gives

$$I_{a,b}(p,q) = \frac{1}{p}\sum_{\ell=0}^{p-1} \chi_{a-1}(u_\ell)\chi_{b-1}(v_\ell). \tag{16}$$

Indeed, expansion produces the phase $\exp\{\pi i\ell[(1+q)r + (1-q)t]/p\}$, and its cyclic average is exactly the congruence in (2).

For $m \geq 1$, define

$$\begin{aligned} D_J^{\mathrm{form}}(m;\ell) = {} & \chi_{m+J-1}(u_\ell)\chi_{m-J-1}(v_\ell) \\ & - \chi_{m-J-1}(u_\ell)\chi_{m+J-1}(v_\ell). \end{aligned}$$

Lemma 5.3 (The even half-turn)**.** *If p is even, then the unique half-turn $\ell = p/2$ has*

$$D_J^{\mathrm{form}}(m;p/2) = 0 \qquad (m \in \mathbb{Z}).$$

Thus it contributes no signed eta term, before analytic continuation.

Proof. Because q is a unit modulo even p, it is odd. Hence at $\ell = p/2$,

$$u_\ell = \frac{\pi(1+q)}{2}, \qquad v_\ell = \frac{\pi(1-q)}{2}$$

are both integral multiples of π. For $\varepsilon = \pm 1$ the central character value is $\chi_h(\varepsilon) = (h+1)\varepsilon^h$. Put $A = m+J-1$, $B = m-J-1$, and let $\varepsilon_u, \varepsilon_v$ be the two central signs. Since $A - B = 2J$,

$$\frac{\varepsilon_u^A \varepsilon_v^B}{\varepsilon_u^B \varepsilon_v^A} = \left(\frac{\varepsilon_u}{\varepsilon_v}\right)^{2J} = 1.$$

The common amplitude is $(A+1)(B+1)$, so exchanging the two indices leaves the character product unchanged, including the extended-character cases. Their signed difference is zero. □

The identity element contributes zero. Product-to-sum gives

$$D_J^{\mathrm{form}}(m;\ell) = \frac{\sin(2m\beta_\ell)\sin(2J\alpha_\ell) - \sin(2m\alpha_\ell)\sin(2J\beta_\ell)}{\sin(\alpha_\ell+\beta_\ell)\sin(\beta_\ell-\alpha_\ell)}. \tag{17}$$

At a vanishing denominator this expression means the removable character limit in the preceding definition.

For $\Re s$ sufficiently large, define the formal Dirichlet series

$$\eta_J^{\mathrm{form}}(s) = \frac{1}{p}\sum_{\ell=1}^{p-1}\sum_{m\geq 1} D_J^{\mathrm{form}}(m;\ell)m^{-s}.$$

Lemma 5.4 (Abel and eta continuation)**.** *Write $\eta_J^{\mathrm{form}} := \eta_J^{\mathrm{form}}(0)$ for the value at zero of the analytically continued formal eta series. This value equals its Abel value:*

$$\eta_J^{\mathrm{form}} = \frac{1}{2p}\sum_{\ell=1}^{p-1}\frac{\sin(2J\alpha_\ell)\cot\beta_\ell - \sin(2J\beta_\ell)\cot\alpha_\ell}{\sin(\alpha_\ell+\beta_\ell)\sin(\beta_\ell-\alpha_\ell)}. \tag{18}$$

For even p, the summand at $\ell = p/2$ is understood through the character definition and is zero by Lemma 5.3.

Proof. For $\theta \notin \pi\mathbb{Z}$,

$$\lim_{t\uparrow 1}\sum_{m\geq 1} t^m \sin(2m\theta) = \lim_{t\uparrow 1}\frac{t\sin(2\theta)}{1-2t\cos(2\theta)+t^2} = \frac{1}{2}\cot\theta.$$

This proves the formula when both character denominators are nonzero. For fixed ℓ, a removable character limit is a finite linear combination of $m^\nu\omega^m$, with $\nu \leq 1$ and ω a nontrivial $2p$th root of unity. Here is the denominator classification explicitly. One has

$$\sin u_\ell = 0 \iff p \mid (q+1)\ell, \qquad \sin v_\ell = 0 \iff p \mid (q-1)\ell.$$

If both vanish, adding and subtracting the congruences gives $p \mid 2\ell$. Since $1 \leq \ell < p$, this is possible exactly when p is even and $\ell = p/2$; the resulting signed character is identically zero by Lemma 5.3. If exactly one denominator vanishes, taking the character limit differentiates only a finite exponential sum and hence adds at most one factor of m. Its m-dependent roots are among $e^{\pm 2i\alpha_\ell}$ and $e^{\pm 2i\beta_\ell}$. The latter are nontrivial because $0 < \ell < p$, and the former are nontrivial because q is a unit modulo p. Thus $\omega \neq 1$ in every remaining removable case, and

$$\sum_{m\geq 1}^{\mathrm{Abel}} \omega^m = \mathrm{Li}_0(\omega), \qquad \sum_{m\geq 1}^{\mathrm{Abel}} m\omega^m = \mathrm{Li}_{-1}(\omega).$$

These are also the values at $s = 0$ of the corresponding continued Dirichlet series, so Abel and eta continuation agree in every removable case. □

If

$$x = \cos(2\alpha_\ell), \quad y = \cos(2\beta_\ell), \quad f_J(x) = (1-x)U_{J-1}(x),$$

then

$$\sin(2J\alpha_\ell) = \cot\alpha_\ell\, f_J(x), \qquad \sin(\alpha_\ell+\beta_\ell)\sin(\beta_\ell-\alpha_\ell) = \frac{x-y}{2}.$$

Consequently

$$\eta_J^{\text{form}} = \frac{1}{p}\sum_{\ell=1}^{p-1} \cot\alpha_\ell \cot\beta_\ell \, P_J(x,y), \qquad P_J(x,y) = \frac{f_J(x) - f_J(y)}{x-y}. \tag{19}$$

At $x = y$ we set $P_J(x,x) = f_J'(x)$. This polynomial continuation includes $q = \pm 1$ and composite singular group elements without division by zero.

Lemma 5.5 (All-rank generating function)**.** *The divided differences satisfy*

$$\sum_{J\geq 1} P_J(x,y) z^J = -\frac{z(1-z)^2}{(1-2xz+z^2)(1-2yz+z^2)}. \tag{20}$$

Equivalently,

$$\frac{U_n(x) - U_n(y)}{x-y} = 2\sum_{a=0}^{n-1} U_a(x) U_{n-1-a}(y).$$

Proof. Insert the standard generating function $\sum_{n\geq 0} U_n(x) z^n = (1-2xz+z^2)^{-1}$ and simplify the divided difference. The coefficient identity follows by comparing powers of z. □

Lemma 5.6 (Exact trigonometric decomposition)**.** *For angles at which the cotangents are defined,*

$$\begin{aligned} &\cot A \cot B \, P_J(\cos 2A, \cos 2B) \\ =\ & -J \cot A \cot B \\ & + 2\sum_{r=1}^{J-1} (J-r)\left[\sin(2rA)\cot B + \sin(2rB)\cot A\right] \\ & - 4 \sum_{\substack{r,s\geq 1 \\ r+s\leq J-1}} (J-r-s)\sin(2rA)\sin(2sB). \end{aligned} \tag{21}$$

Proof. Assume first that both cotangents are nonzero. Let

$$D_A = 1 - 2z\cos 2A + z^2, \qquad D_B = 1 - 2z\cos 2B + z^2.$$

The generating function of the right side is

$$\begin{aligned} & -\frac{z}{(1-z)^2}\cot A \cot B \\ & + \frac{2z^2}{(1-z)^2}\left[\frac{\sin 2A \cot B}{D_A} + \frac{\sin 2B \cot A}{D_B}\right] - \frac{4z^3 \sin 2A \sin 2B}{(1-z)^2 D_A D_B}. \end{aligned}$$

Put the terms over a common denominator and use $\sin 2A/\cot A = 2\sin^2 A$ together with $D_A = (1-z)^2 + 4z\sin^2 A$, and similarly for B. The numerator becomes $-z(1-z)^4 \cot A \cot B$. After cancelling $(1-z)^2$, the result is

$$-\frac{z(1-z)^2 \cot A \cot B}{D_A D_B},$$

which is $\cot A \cot B$ times (20). Equality of coefficients proves the lemma. Both sides of (21) are continuous at angles where a cotangent vanishes, so the identity extends to those cases by continuity. □

Lemma 5.7 (Finite Fourier identities)**.** *For all integers k, u, v,*

$$\frac{1}{p}\sum_{\ell=1}^{p-1}\cot\alpha_\ell\cot\beta_\ell = 4s(q,p), \tag{22}$$

$$\sum_{\ell=1}^{p-1}\sin\left(\frac{2\pi k\ell}{p}\right)\cot\left(\frac{\pi\ell}{p}\right) = \rho_p(k), \tag{23}$$

$$\frac{1}{p}\sum_{\ell=1}^{p-1}\sin\left(\frac{2\pi u\ell}{p}\right)\sin\left(\frac{2\pi v\ell}{p}\right) = \frac{1}{2}\left[\delta_p(u-v)-\delta_p(u+v)\right]. \tag{24}$$

The missing factor $1/p$ in (23) is intentional: $\rho_p(k)$ is the unnormalized finite sine–cotangent sum.

Proof. Equation (22) is the classical cotangent expression for the Dedekind sum, after permuting nonzero residues by q. If $1 \le k < p$, then

$$\sin(2kx)\cot x = 1 + \cos(2kx) + 2\sum_{a=1}^{k-1}\cos(2ax).$$

Summing at $x = \pi\ell/p$ gives $p - 2k$; reducing an arbitrary k modulo p proves (23). Product-to-sum and finite exponential orthogonality give (24). □

Proof of Theorem 5.1. Apply Lemma 5.6 to (19) and average over ℓ. The first term is $-4Js(q,p)$ by (22). The first sine–cotangent term gives $\rho_p(rq)$ by (23). In the second, the change of variables $h \equiv q\ell \pmod p$ gives

$$\sum_{\ell=1}^{p-1}\sin(2r\beta_\ell)\cot\alpha_\ell = \rho_p(rq^*).$$

The sine–sine terms give, by (24),

$$-2\sum_{\substack{r,s\ge1\\ r+s\le J-1}}(J-r-s)\left[\delta_p(rq-s)-\delta_p(rq+s)\right].$$

Thus the first three lines of (15) equal the formal eta value. The modular indicators arise from finite sine orthogonality, not only from degenerate character denominators.

These finite Fourier identities hold for every modulus $p \ge 2$ and use no inverse of 2 modulo p. When p is even, the half-turn has $\alpha_{p/2} = \pi q/2$ and $\beta_{p/2} = \pi/2$. Hence its cotangents and all the sine factors appearing in Lemma 5.6 vanish; this is the zero character contribution already identified in Lemma 5.3. Reinstating that zero term therefore gives the same Dedekind, residue, and modular-indicator expressions with no extra even-level correction.

For $m \ge J+1$, (16) identifies the averaged formal character with the physical signed multiplicity at $n = m - J - 1$. At $m = J$ it vanishes because $\chi_{-1} = 0$. If $1 \le m < J$, then

$$\chi_{m-J-1} = -\chi_{J-m-1},$$

so the averaged formal multiplicity is

$$I_{J-m,J+m}(p,q) - I_{J+m,J-m}(p,q).$$

The physical spectrum begins at $m = J+1$. Its eta value is therefore the formal value minus the sum of these sub-threshold terms, namely $L_J(p,q)$. This gives the final line of (15) with the asserted sign. □

The logical distinction between the formal character device and the geometric spectrum can be summarized as

$$\boxed{\begin{array}{c}\text{extended-character series} \longrightarrow \text{Abel/Hurwitz continuation} \longrightarrow \text{finite Fourier evaluation} \\ \longrightarrow \text{subtract } L_J = \text{physical TT eta}\end{array}} \tag{25}$$

The first three objects include formal sub-threshold indices; only the final subtraction imposes the lower boundary $n = 0$ of (6).

Corollary 5.8 (Rank-one control)**.** *At $J = 1$ all correction sums vanish and*

$$\eta_1(L(p,q)) = -4s(q,p). \tag{26}$$

This is the classical odd-signature or curl eta value in the orientation of Section 2.

Corollary 5.9 (Even-order anchor)**.** *For every integer $J \geq 1$,*

$$\eta_J(L(2,1)) = \eta_J(\mathbb{RP}^3) = 0.$$

Proof. Modulo 2 one has $-1 \equiv 1$. The orientation identity (4) therefore gives $\eta_J(L(2,1)) = -\eta_J(L(2,1))$. □

Corollary 5.10 (Rank-two specialization)**.** *At $J = 2$,*

$$\eta_2(L(p,q)) = -8s(q,p) + \frac{2}{p}\left[\rho_p(q) + \rho_p(q^*)\right] - L_2(p,q). \tag{27}$$

In particular,

$$\eta_2(L(p,1)) = -\frac{2(p^2 - 6p + 14)}{3p} \qquad (p \geq 3).$$

The remaining case $p = 2$ is the zero value of Corollary 5.9.

Example 5.11 ($J = 2$ on $L(5,2)$)**.** Here $q^* = 3 \equiv -2 \pmod 5$. Inversion and oddness of the Dedekind sum give $s(2,5) = s(3,5) = -s(2,5) = 0$. Moreover

$$\rho_5(2) = 1, \qquad \rho_5(3) = -1.$$

The sole low-mode term is

$$L_2(5,2) = I_{1,3}(5,2) - I_{3,1}(5,2) = 1 - 1 = 0 :$$

in each count only the zero weight satisfies the congruence. Thus every piece of (27) is visible and

$$\eta_2(L(5,2)) = 0.$$

This example also illustrates why the low-mode term must be evaluated even when it happens to cancel.

6 Homogeneous and rank-symmetry corollaries

Theorem 6.1 (Homogeneous lens spaces)**.** *For every integer $J \geq 1$ and $p \geq \max(2, 2J-1)$,*

$$\eta_J(L(p,1)) = -\frac{J\,(p^2 - 3Jp + 4J^2 - 2)}{3p}. \tag{28}$$

For $J \geq 2$ the threshold is sharp. At the first even level below it, $p = 2J-2$, the exact value exceeds the stable expression by 2; for $J \geq 3$, at the first odd level below it, $p = 2J-3$, the excess is 4.

Proof. Specialize (15) at $q = q^* = 1$. Direct summation of the sawtooth definition gives

$$s(1,p) = \frac{(p-1)(p-2)}{12p},$$

so the Dedekind term is $-J(p^2 - 3p + 2)/(3p)$. Since $1 \leq a \leq J-1 < p$, one has $\rho_p(a) = p - 2a$, and therefore

$$\frac{2}{p}\sum_{a=1}^{J-1}(J-a)\,2\rho_p(a) = 2J(J-1) - \frac{4J(J^2-1)}{3p}. \tag{29}$$

In the triangular correction, $|a-b| < p$ and $0 < a+b < p$. Thus $\delta_p(a-b) = \mathbf{1}_{a=b}$ and $\delta_p(a+b) = 0$, and its value is

$$-2\sum_{a=1}^{\lfloor (J-1)/2 \rfloor}(J-2a) = \begin{cases} -J(J-2)/2, & J \text{ even}, \\ -(J-1)^2/2, & J \text{ odd}. \end{cases} \tag{30}$$

It remains to evaluate the low-mode term rather than assume it is stable. For $q = 1$, the congruence in (2) is $r \equiv 0 \pmod{p}$. Every weight occurring there has absolute value at most $2J - 2 < p$, so only $r = 0$ is possible. It occurs precisely when the first index is odd, whence

$$I_{a,b}(p,1) = b\mathbf{1}_{a \text{ odd}}, \qquad L_J(p,1) = \sum_{\substack{1 \leq m < J \\ J-m \text{ odd}}} 2m = \left\lfloor \frac{J^2}{2} \right\rfloor.$$

Combining this with (30), the last two lines of the master formula contribute $-J(J-1)$ for either parity of J. Adding this to the Dedekind term and (29) yields

$$\eta_J(L(p,1)) = -\frac{J(p^2 - 3p + 2)}{3p} + J(J-1) - \frac{4J(J^2-1)}{3p},$$

which is (28).

For the first even level below the stable range, let $J \geq 2$ and $p = 2J-2$. The only new low-mode event occurs at $m = J-1$. The first factor in $I_{2J-1,1}(p,1)$ now admits the three weights $r = -p, 0, p$ rather than only $r = 0$, while $I_{1,2J-1}(p,1)$ is unchanged. Consequently the exact L_J is smaller than its stable value by 2, and the exact eta invariant exceeds (28) by 2.

Finally let $J \geq 3$ and $p = 2J-3$. The preceding Dedekind, residue, and triangular calculations remain valid. In the low-mode count, however, the weights $r = \pm p$ first enter, and only for $m = J-2$. More explicitly,

$$I_{2,2J-2}(p,1) = 0, \qquad I_{2J-2,2}(p,1) = 4.$$

Thus the exact L_J is smaller than its stable value by 4, so the exact eta invariant exceeds the polynomial in (28) by 4. This proves the asserted sharpness. Notice that no primality assumption was used. □

As a fixed worked example, we use the same level 137 as the companion coexact-mode calculation [Mah26b]. The first five ranks at $L(137,1)$ are

$$(\eta_1,\dots,\eta_5) = -\frac{1}{137}(6120, 11974, 17570, 22916, 28020). \tag{31}$$

Equation (28) holds at every level in its stated stable range.

Theorem 6.2 (Nonsingular rank symmetries)**.** *Suppose*

$$\gcd(1+q,p) = \gcd(1-q,p) = 1.$$

Then

$$\eta_{J+p}(L(p,q)) = \eta_J(L(p,q)). \tag{32}$$

For $1 \le J \le p-1$,

$$\eta_{p-J}(L(p,q)) = -\eta_J(L(p,q)), \qquad \eta_p(L(p,q)) = 0. \tag{33}$$

Proof. Under the stated hypothesis the denominators in (17) are nonzero. Let F_J denote the formal Abel value in (18), and set

$$K_J(m) = \frac{1}{p}\sum_{\ell=1}^{p-1} \frac{\sin(2m\beta_\ell)\sin(2J\alpha_\ell) - \sin(2m\alpha_\ell)\sin(2J\beta_\ell)}{\sin(\alpha_\ell+\beta_\ell)\sin(\beta_\ell-\alpha_\ell)}. \tag{34}$$

The proof of Theorem 5.1 shows that

$$\eta_J = F_J - c_J, \qquad c_J = \sum_{m=1}^{J} K_J(m), \tag{35}$$

where including $m = J$ is harmless because $K_J(J) = 0$.

The elementary sine identities in the rank variable give

$$F_{J+p} = F_J, \qquad F_{p-J} = -F_J, \qquad F_p = 0.$$

The same identities applied to (34) give

$$\begin{aligned} K_{J+p}(m) &= K_J(m), & K_{p-J}(m) &= -K_J(m), & K_p(m) &= 0, \\ K_J(m+p) &= K_J(m), & K_J(p-m) &= -K_J(m), & K_J(0) &= K_J(J) = 0. \end{aligned}$$

Consequently a complete period in m sums to zero, and hence

$$c_{J+p} = c_J, \qquad c_p = 0.$$

For reflection, use the vanishing of a full period and $K_J(p-m) = -K_J(m)$:

$$\sum_{m=1}^{p-J} K_J(m) = -\sum_{m=p-J+1}^{p} K_J(m) = \sum_{n=0}^{J-1} K_J(n) = \sum_{n=1}^{J-1} K_J(n) = c_J.$$

Therefore $c_{p-J} = -c_J$ for every $1 \le J \le p-1$, including the upper half of the fundamental rank range. Substitution in (35) proves (32) and (33). □

Remark 6.3. The nonsingularity hypothesis has nonempty scope exactly when $\gcd(p,6) = 1$. If $2 \mid p$, every unit q is odd, so both $1+q$ and $1-q$ share the factor 2 with p. If $3 \mid p$, every unit is congruent to $\pm 1 \pmod 3$, so one of $1+q$ and $1-q$ shares the factor 3 with p. Conversely, if $\gcd(p,6) = 1$, then $q = 2$ is a unit modulo p and satisfies both gcd conditions. Thus this theorem has nonempty scope precisely at levels coprime to 6, even though Theorem 5.1 holds for all $p \ge 2$. These are arithmetic symmetries of the rank-indexed eta sequence, not a claim of periodic physical spin.

7 Four infinite eta-zero families

Theorem 7.1 (Arithmetic zero families)**.** *Let $J \geq 2$ and $p_J = 2J^2 - 1$. Then*

$$\begin{aligned} \eta_J(L(p_J, J)) &= 0, & \eta_J(L(p_J, J+1)) &= 0, \\ \eta_J(L(p_J, 2J-1)) &= 0, & \eta_J(L(p_J, J(J-1))) &= 0. \end{aligned} \tag{36}$$

Their images under $q \mapsto q^{-1}$ and $q \mapsto -q$ are also zeros. Every level p_J in these families is odd. The four displayed formulae need not represent four distinct lens orbits at small J.

Proof. Write $p = p_J$. The identities

$$\begin{aligned} J(2J) &= p + 1, \\ (J+1)(p - 2J + 2) &\equiv 1 \pmod{p}, \\ (2J-1)(2J+1) &= 2p + 1, \\ (4J+2)J(J-1) &= (2J-1)p - 1 \end{aligned}$$

show that all four displayed residues are units and give the inverses used below. Define the residue contribution

$$R_J(q) = \frac{2}{p} \sum_{a=1}^{J-1} (J-a)[\rho_p(aq) + \rho_p(aq^*)].$$

For $q = J, J+1, 2J-1$, repeated Dedekind reciprocity gives, respectively,

$$s(J, p) = \frac{(J-1)(J^2 - J - 1)}{2p}, \tag{37}$$

$$s(J+1, p) = \frac{(J-3)(J-1)(J+1)}{6p}, \tag{38}$$

$$s(2J-1, p) = \frac{(J-1)(2J^2 - 4J - 3)}{6p}. \tag{39}$$

Their inverse residues are $2J$, $p - 2J + 2$, and $2J + 1$. No residue wraps except the explicitly reduced second member of the middle pair, so for $1 \leq a < J$,

$$\begin{aligned} \rho_p(aJ) + \rho_p(2aJ) &= 2p - 6Ja, \\ \rho_p(a(J+1)) + \rho_p(a(p - 2J + 2)) &= 2a(J-3), \\ \rho_p(a(2J-1)) + \rho_p(a(2J+1)) &= 2p - 8Ja. \end{aligned}$$

Using $\sum_{a=1}^{J-1}(J-a) = J(J-1)/2$ and $\sum_{a=1}^{J-1} a(J-a) = J(J-1)(J+1)/6$ now gives

$$R_J(q) = 4J\, s(q, p) \qquad (q = J, J+1, 2J-1). \tag{40}$$

The remaining two master-formula terms vanish for these three choices. Indeed, whenever $a, b \geq 1$ and $a + b \leq J - 1$, one has

$$0 < aq - b < aq + b < p,$$

so the triangular δ_p sum is zero. For the low-mode term put $u = (r - t)/2$ and $v = (r + t)/2$. The two weights have the same parity, so $u, v \in \mathbb{Z}$, $|u|, |v| \leq J - 1$, and the congruence in (2) becomes

$$v + qu \equiv 0 \pmod{p}.$$

For each of the three q values, $|v+qu| < p$ and $q > J-1$; hence $u = v = 0$. This zero-weight solution occurs symmetrically in the exchanged counts, so $L_J(p,q) = 0$. Equations (37)–(40) therefore prove the first three zeros.

For the fourth family use $q_0 = 4J+2$. Its least positive inverse is

$$q_0^* = p - J(J-1) = J^2 + J - 1,$$

and $J(J-1) \equiv -q_0^* \pmod{p}$. Set $h = \lfloor (J-1)/2 \rfloor$. For $1 \le a < J$, direct reduction gives

$$\langle aq_0 \rangle_p = \begin{cases} aq_0, & a \le h, \\ aq_0 - p, & a > h, \end{cases}$$

$$\langle aq_0^* \rangle_p = \begin{cases} a(2J-1)/2, & a \text{ even}, \\ [p + a(2J-1)]/2, & a \text{ odd}. \end{cases}$$

Substitution into the finite weighted sum defining R_J and evaluation of the resulting arithmetic sums yields the following expression. Explicitly,

$$\rho_p(aq_0) + \rho_p(aq_0^*) = 2p - a(10J+3) + 2p\mathbf{1}_{a>h} - p\mathbf{1}_{a \text{ odd}},$$

and the evaluation uses

$$\sum_{a=1}^{J-1}(J-a) = \frac{J(J-1)}{2}, \qquad \sum_{a=1}^{J-1} a(J-a) = \frac{J(J-1)(J+1)}{6},$$

$$\sum_{h<a<J}(J-a) = \frac{H(H+1)}{2}, \qquad H = \left\lfloor \frac{J}{2} \right\rfloor, \qquad \sum_{\substack{1 \le a < J \\ a \text{ odd}}} (J-a) = \begin{cases} H^2, & J = 2H, \\ H(H+1), & J = 2H+1. \end{cases}$$

Consequently

$$R_J(q_0) = \begin{cases} \dfrac{J(2J^3 - 9J^2 + 4J + 6)}{3p}, & J \text{ even}, \\ \dfrac{J(J-1)(2J^2 - 13J - 9)}{3p}, & J \text{ odd}. \end{cases} \tag{41}$$

The Euclidean remainder chains

$$\begin{aligned} p &\equiv J \pmod{q_0}, \quad q_0 \equiv 2 \pmod{J}, && J \text{ odd}, \\ p &\equiv -(J+1) \pmod{q_0}, \quad q_0 \equiv -2 \pmod{J+1}, && J \text{ even}, \end{aligned}$$

together with Dedekind reciprocity and $s(-a,b) = -s(a,b)$ give

$$s(q_0,p) = \begin{cases} \dfrac{2J^3 - 9J^2 + 4J + 6}{12p}, & J \text{ even}, \\ \dfrac{(J-1)(2J^2 - 13J - 9)}{12p}, & J \text{ odd}. \end{cases} \tag{42}$$

Thus (41) is exactly $4J$ times (42).

It remains only to exclude corrections. The least absolute residue of p modulo q_0 is J for odd J and $J+1$ for even J. In the triangular range, $0 < aq_0 \pm b < 2p$ and $|b| \le J-2$; divisibility by p would force $b \equiv \pm p \pmod{q_0}$, contradicting that least-residue bound. Hence the δ_p term is zero. In the low-mode range the same variables u, v satisfy $|u|, |v| \le J-1$ and $|v + q_0 u| < 2p$. The values $\pm p$ are again excluded by the least-residue bound, while $v + q_0 u = 0$ forces $u = v = 0$. The surviving zero weight is symmetric, so $L_J(p, q_0) = 0$. Therefore $\eta_J(L(p,q_0)) = 0$, and (4) transfers this zero to $q = J(J-1)$ and to every stated orbit image. □

A complete classification of zero orbits beyond the four constructions above and the orientation-forced condition $q^2 \equiv -1 \pmod{p}$ remains an open arithmetic problem.

8 Computational verification

Exact computations were used to test the projector, Hurwitz-zeta, Dedekind–Rademacher, and compact-residue forms of the answer across prime, composite, denominator-singular, and sharp-boundary cases. Exact symbolic fiber matrices were compared through rank $J = 10$, and independent weight-space calculations tested the resulting formulas through rank $J = 64$. Separate checks covered the even half-turn, the $\mathbb{RP}^3$ anchor, and the four zero families. A companion data set contains 16,177 exact rank-one spectral-flow rows and higher-rank regression values. All comparisons use rational or integer arithmetic; they test the formulas and conventions but are not substitutes for the proofs.

9 Relation to antecedents and scope

Homma supplies the higher-spin self-gradient, its signed sphere branches, and the constant-curvature TT preservation identity [Hom01, HT21]. David–Gaberdiel–Gopakumar supply the arbitrary-rank TT representations and quotient character machinery [DGG10]. Bär and APS provide the rank-one curl or odd-signature control [Bär19, APS75]. In rank two, Castro–Lashkari–Maloney give the direct operator and explicit unresolved phase gap [CLM11]. Basu gives unsigned spin-three and spin-four lens-space determinant expressions and conjectures the pattern above spin four [Bas15]. Degeratu's Molien formula evaluates twisted Dirac eta invariants, not the projected TT tensor-curl family [Deg09]. Donnelly's equivariant calculation for the APS boundary signature operator, as restated by Szczepański as Propositions 1–2 (explicitly attributed there to Donnelly's Propositions 4.6–4.7) [Szc12, Sec. 2, Props. 1–2], gives a cotangent fixed-point formula for a special class of affine torus actions [Don78]. This is a method antecedent for the equivariant trigonometric route. Those propositions neither evaluate nor exclude the projected TT tensor-curl family considered here. Capoferri–Vassiliev give a recent microlocal construction of curl spectral asymmetry through regularized pseudodifferential projections [CV25]. This is a close analytic antecedent at $J = 1$; it does not contain the arbitrary-rank lens-space evaluation here. The companion boundary analysis, Dirac-to-tensor-curl transfer, and bivariate McKay–Chebyshev algebra are developed in [Mah26d, Mah26c, Mah26a].

Thus no novelty is attached to the operator, its sphere spectrum, cyclic projection, or the general equivariant method. The explicit signed evaluation (15), its composite singular resolution, and the corollaries of Sections 6 and 7 are, to the best of our knowledge, not present in the antecedent literature. This is a bounded historical statement rather than a claim of an exhaustive search.

The present result is a spectral theorem for all nontrivial round lens spaces. It does not assemble the auxiliary eta and kernel terms of a full APS correction, establish an off-round reducing split, or determine a complete one-loop phase with its longitudinal, trace, ghost, framing, and counterterm data. Nor does the formula select a lens level or classify every eta-zero triple (p, J, q).

10 Discussion

The formula exhibits a useful separation of structure. The Dedekind term is the classical rank-one spectral asymmetry multiplied by J. The residue term records finite helicity weights, the δ_p term records modular congruence or Fourier-aliasing coincidences, and L_J repairs the finite low-energy boundary of the physical TT spectrum. Removable denominator-singular character limits are handled separately by the polynomial continuation in (19). Prime and composite levels are thereby handled by a single expression.

The most immediate mathematical continuation is a chamber decomposition of the congruence and low-mode corrections, followed by a classification of the remaining zero orbits. The reflection $\eta_{p-J} = -\eta_J$ suggests that the rank sequence should be understood as a finite sine expansion, with the generalized Dedekind data playing the role of Fourier coefficients. The generating kernel (20) similarly packages the rank dependence into a rational function, pointing toward a coefficient-extraction approach to the zero-orbit problem.

Metric variation is also open: the projected operator is analytically well-defined more generally, but the exact lens formula uses round quotient representation theory. Other natural directions include chamber formulas for the congruence corrections and a higher-rank analogue of the rank-one Chern–Simons normalization in Appendix B.

A An equivalent Rademacher-sum form

Define

$$R_{p,q}(c) = \sum_{r=0}^{p-1} \left(\!\left(\frac{r}{p}\right)\!\right) \left(\!\left(\frac{qr+c}{p}\right)\!\right)$$

and, with $S_N = 0$ for $N < 0$,

$$S_N = \sum_{a=0}^{N} \sum_{w \in W_a} \sum_{t \in W_{N-a}} R_{p,q}(wq+t).$$

Proposition A.1. *The eta invariant also has the form*

$$\eta_J = 4(2S_{J-2} - S_{J-1} - S_{J-3}) - L_J. \tag{43}$$

Proof. Let $\zeta = e^{2\pi \mathrm{i}/p}$. The finite Fourier transform of the sawtooth function is

$$\sum_{r=0}^{p-1} \left(\!\left(\frac{r}{p}\right)\!\right) \zeta^{-\ell r} = \frac{\mathrm{i}}{2} \cot \frac{\pi \ell}{p}, \qquad 1 \le \ell < p.$$

Fourier inversion and the substitution $r \mapsto qr$ therefore give

$$4pR_{p,q}(c) = \sum_{\ell=1}^{p-1} \cot \alpha_\ell \cot \beta_\ell \, \zeta^{\ell c}. \tag{44}$$

Since $\sum_{w \in W_a} \zeta^{q\ell w} = U_a(\cos 2\alpha_\ell)$ and $\sum_{t \in W_b} \zeta^{\ell t} = U_b(\cos 2\beta_\ell)$, put

$$C_N(x,y) = \sum_{a=0}^{N} U_a(x) U_{N-a}(y), \qquad C_N = 0 \quad (N < 0).$$

The generating function of the Chebyshev polynomials gives

$$C_N(x,y) = \frac{U_{N+1}(x) - U_{N+1}(y)}{2(x-y)}.$$

Using $U_{n+1} = 2xU_n - U_{n-1}$ in this quotient yields

$$2C_{J-2} - C_{J-1} - C_{J-3} = \frac{(1-x)U_{J-1}(x) - (1-y)U_{J-1}(y)}{x-y} = P_J(x,y). \tag{45}$$

Summing (44) over the weights in S_N and applying (45) gives $4(2S_{J-2} - S_{J-1} - S_{J-3}) = \eta_J^{\mathrm{form}}$ by (19). Finally, the physical spectrum begins at $m = J+1$, so the sub-threshold term L_J is subtracted exactly as in the proof of Theorem 5.1. □

B Rank-one Chern–Simons normalization

This appendix compares (26) with Jeffrey's spectral-flow normalization; it is not a premise of Theorem 5.1. Define

$$R_m(p,q) = \frac{1}{p}\sum_{k=1}^{p-1} \cos\left(\frac{2\pi mk}{p}\right)\cot\left(\frac{\pi k}{p}\right)\cot\left(\frac{\pi qk}{p}\right). \tag{46}$$

Jeffrey's cotangent formula gives

$$R_0(p,q) = 4s(q,p) = -\eta_1(L(p,q)).$$

Proposition B.1 (Jeffrey spectral-flow control)**.** *Assume $p \geq 3$ is odd and $1 \leq n \leq (p-1)/2$. Let I_n be the spectral flow between the trivial connection and the nth nontrivial flat $SU(2)$ connection in Jeffrey's convention. Then*

$$I_n = \frac{8q^*n^2}{p} - 2 + R_0(p,q) - R_{2n}(p,q) = \frac{8q^*n^2}{p} - 2 - \eta_1(L(p,q)) - R_{2n}(p,q). \tag{47}$$

Proof. Proposition 5.2 of [Jef92] states

$$I_n + 2 = \frac{8q^*n^2}{p} + \frac{2}{p}\sum_{k=1}^{p-1} \cot\left(\frac{\pi k}{p}\right)\cot\left(\frac{\pi qk}{p}\right)\sin^2\left(\frac{2\pi nk}{p}\right).$$

Use $2\sin^2 x = 1 - \cos 2x$ and (46). The inverse q^* is essential. □

The finite Fourier identity

$$R_m(p,q) = 4\sum_{r=0}^{p-1} \left(\!\left(\frac{r}{p}\right)\!\right)\left(\!\left(\frac{qr+m}{p}\right)\!\right),$$

gives an exact rational evaluation of R_m. Together with $R_0 = 4s$ and $\eta_1 = -4s$, it fixes the sign and normalization in (47).

At Chern–Simons level κ, Jeffrey's canonical two-framing invariant also contains

$$\exp\left(\frac{2\pi i}{4\kappa}\,12s(q,p)\right) = \exp\left(\frac{2\pi i}{4\kappa}\left(-3\eta_1(L(p,q))\right)\right).$$

Her comparison of the stationary-phase expression with the exact invariant is asymptotic rather than exact at finite κ. Accordingly, this comparison fixes the rank-one convention but does not supply a $J \geq 2$ Chern–Simons phase.

Acknowledgment of assistance

AI-assisted research, computation, checking, and editorial support used OpenAI Codex and Anthropic Claude. Luqman Mahmood directed the work and is responsible for the claims and final text. These internal checks are not external peer review.

References

[APS75] Michael F. Atiyah, Vijay K. Patodi, and Isadore M. Singer. Spectral asymmetry and riemannian geometry. II. *Mathematical Proceedings of the Cambridge Philosophical Society*, 78(3):405–432, 1975.

[Bär19] Christian Bär. The curl operator on odd-dimensional manifolds. *Journal of Mathematical Physics*, 60(3):031501, 2019.

[Bas15] Rudranil Basu. Higher spin de sitter quantum gravity. *Journal of High Energy Physics*, 2015(10):151, 2015.

[CLM11] Alejandra Castro, Nima Lashkari, and Alexander Maloney. Quantum topologically massive gravity in de sitter space. *Journal of High Energy Physics*, 2011(8):040, 2011.

[CV25] Matteo Capoferri and Dmitri Vassiliev. A microlocal pathway to spectral asymmetry: Curl and the eta invariant. *arXiv preprint*, 2025.

[Deg09] Anda Degeratu. Eta-invariants from molien series. *The Quarterly Journal of Mathematics*, 60(3):303–311, 2009.

[DGG10] Justin R. David, Matthias R. Gaberdiel, and Rajesh Gopakumar. The heat kernel on AdS_3 and its applications. *Journal of High Energy Physics*, 2010(4):125, 2010.

[Don78] Harold Donnelly. Eta invariants for G-spaces. *Indiana University Mathematics Journal*, 27(6):889–918, 1978.

[GL21] Sébastien Gouëzel and Thibault Lefeuvre. Classical and microlocal analysis of the x-ray transform on anosov manifolds. *Analysis & PDE*, 14(1):301–322, 2021.

[Hom01] Yasushi Homma. The higher spin dirac operators on 3-dimensional manifolds. *Tokyo Journal of Mathematics*, 24(2):579–596, 2001.

[HT21] Yasushi Homma and Takuma Tomihisa. The spinor and tensor fields with higher spin on spaces of constant curvature. *Annals of Global Analysis and Geometry*, 60(4):829–861, 2021.

[Jef92] Lisa C. Jeffrey. Chern–simons–witten invariants of lens spaces and torus bundles, and the semiclassical approximation. *Communications in Mathematical Physics*, 147(3):563–604, 1992.

[Mah26a] Luqman Mahmood. Bivariate McKay recurrences and meromorphic tensor-curl spectral functions on spherical space forms. Companion preprint. https://doi.org/10.5281/zenodo.21433689, 2026.

[Mah26b] Luqman Mahmood. Coexact spectral hydrodynamics on L(137,1): Exact modes, triad couplings, and decay. Companion preprint. https://doi.org/10.5281/zenodo.22917335, 2026.

[Mah26c] Luqman Mahmood. Dirac-to-Tensor-Curl Eta Transfer on Round Spherical Space Forms with Affine-ADE Specializations. Companion preprint. https://doi.org/10.5281/zenodo.21433687, 2026.

[Mah26d] Luqman Mahmood. Square elliptic completions and APS boundary information on round spherical quotients. Companion preprint. https://doi.org/10.5281/zenodo.22917346, 2026.

[Pon06] Raphaël Ponge. Spectral asymmetry, zeta functions and the noncommutative residue. *International Journal of Mathematics*, 17(9):1065–1090, 2006.

[Sei15] Jörg Seiler. Parameter-dependent pseudodifferential operators of toeplitz type. *Annali di Matematica Pura ed Applicata*, 194(1):145–165, 2015.

[SS99] A. Yu. Savin and B. Yu. Sternin. Elliptic operators in even subspaces. *Sbornik: Mathematics*, 190(8):1195–1228, 1999.

[SS13] Jörg Seiler and Alexander Strohmaier. On the noncommutative residue for projective pseudodifferential operators. *Journal of Differential Geometry*, 94(3):505–519, 2013.

[Szc12] Andrzej Szczepański. Eta invariants for flat manifolds. *Annals of Global Analysis and Geometry*, 41(2):125–138, 2012.